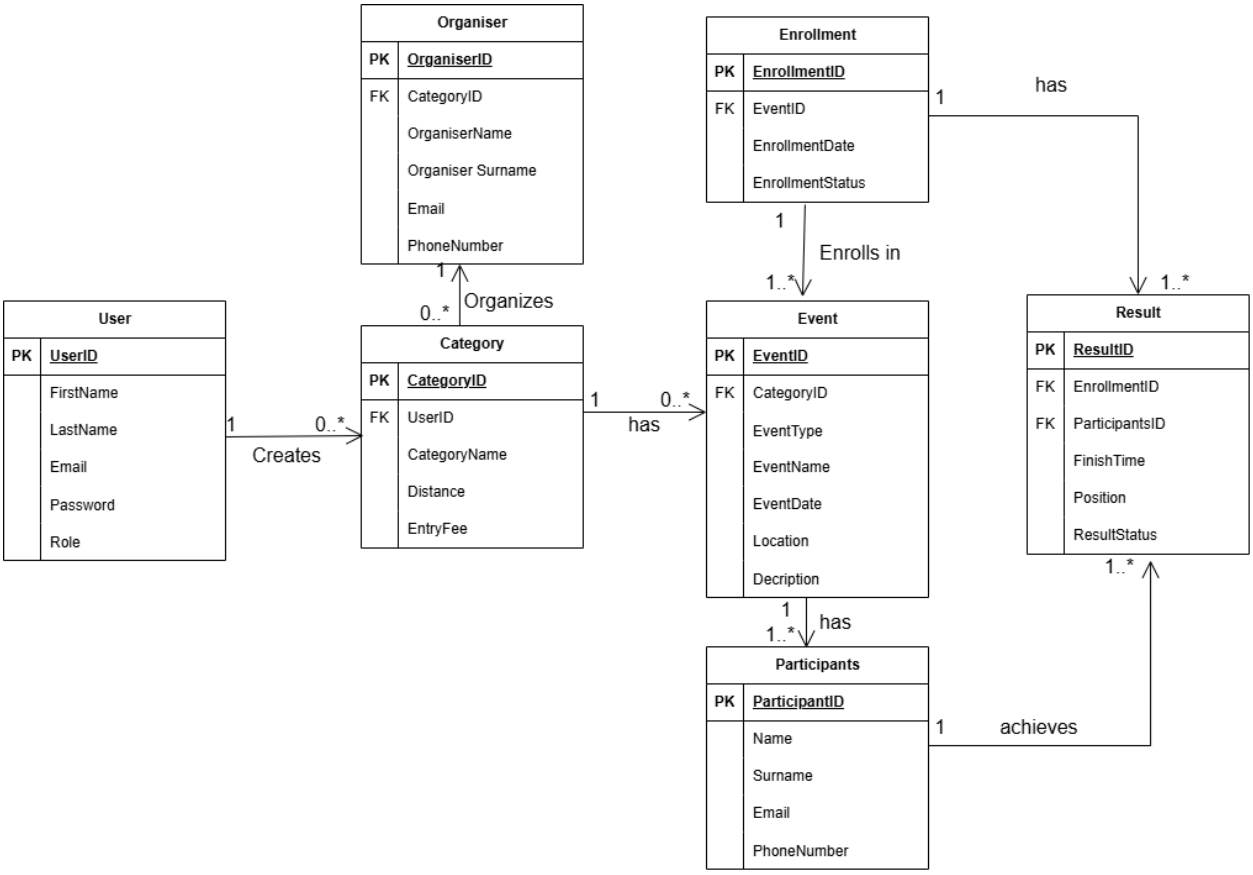
PROG POE PART 1

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Section A: ERD



Business rules

1. User and Category

- A User can create zero or many Categories.

2. Organiser and Category

- An Organiser can organise zero or many Categories.

3. Category and Event

- A Category can have zero or many Events.

4. Event and Enrollment

- An Event can have zero or many Enrollments.

5. Enrollment and Result

- An Enrollment can have one or many Results

6. Event and Participants

- An Event must have one or many Participants

7. Participants and Result

- A Participant can achieve one or many Results

Section B: API Endpoint Plan

1. Authentication

HTTP METHOD	Route	Description	Role required	Request Body	Expected Response
POST	`/api/auth/register`	Registers a new RaceDay user as either an Organiser or Participant.	None (public)	`{ firstName, lastName, email, password, role }`	201 Created - user registered. 400 Bad Request - invalid or missing data. 409 Conflict - email already exists
POST	`/api/auth/login`	Authenticates a registered user and creates an authenticated session containing the user ID and role.	None (public)	`{ email, password }`	200 OK - login successful with authenticated user details/session. 400 Bad Request - invalid request. 401 Unauthorized - incorrect credentials.

User Profile

HTTP METHOD	Route	Description	Role required	Request Body	Expected Response
Get	`/api/users/profile`	Returns the profile information	None (public)	Any (logged in)	200 OK - profile returned.

		belonging to the currently authenticated user.			401 Unauthorized - not authenticated. 404 Not Found - profile does not exist.
PUT	`/api/users/profile`	Updates the currently authenticated user's personal profile information.	Any (logged in)	`{ firstName, lastName, email, phoneNumber }`	200 OK - profile updated. 400 Bad Request - invalid data. 401 Unauthorized - not authenticated. 409 Conflict - email already in use

Events

HTTP METHOD	Route	Description	Role	Request Body	Expected Response
GET	/api/events	Returns available RaceDay events and supports filtering for event browsing.	None (public)	None	200 OK – list of events
GET	/api/events/{eventId	Returns the details of a specific event,	None (public)	None	200 OK – event details. 404 Not Found –

		including its category information			event does not exist
POST	/api/events	Creates a new event owned by the authenticated Organiser.	Organiser	{ categoryId, eventName, eventDate, location, description, eventType, distance, status }	201 Created – event created. 400 Bad Request – invalid data/category. 401 Unauthorized – not authenticated. 403 Forbidden – user is not an Organiser.
PUT	api/events/{eventId	Updates an event owned by the authenticated Organiser.	Organiser	categoryId, eventName, eventDate, location, description, eventType, distance, status }	200 OK – event updated. 400 Bad Request – invalid data. 401 Unauthorized – not authenticated. 403 Forbidden – organiser does not own the event. 404 Not Found – event does not exist
DELETE	api/events/{eventId	Deletes an event owned by the authenticated Organiser when no business rules prevent deletion	Organiser	None	204 No Content – event deleted. 401 Unauthorized – not authenticated. 403 Forbidden – organiser does not own the event. 404 Not Found – event does not exist. 409 Conflict –

					related enrolments/results prevent deletion.

Categories

HTTP METHOD	Route	Description	Role	Request Body	Expected Response
GET	`/api/categories`	Returns all available race categories, including category name, distance, and entry fee.	None (public)	None	200 OK - list of categories.
GET	`/api/categories/{categoryId}`	Returns details of a specific race category.	None (public)	None	200 OK - category details. 404 Not Found - category does not exist.
POST	`/api/categories`	Creates a new age or distance category that can be assigned to an event.	Organiser	`{ categoryName, distance, entryFee }`	201 Created - category created. 400 Bad Request - invalid data. 403 Forbidden - user

					is not an Organiser. 409 Conflict - duplicate category
PUT	/api/categories/{categoryId}	Updates an existing race category.	Organiser	{ categoryName, distance, entryFee }	200 OK - category updated. 400 Bad Request - invalid data. 403 Forbidden - insufficient role. 404 Not Found - category does not exist.
DELETE	/api/categories/{categoryId}	Deletes a category that is not required by an existing event.	Organiser	None	204 No Content - category deleted. 403 Forbidden - insufficient role. 404 Not Found - category does not exist. 409 Conflict -

					category is in use by an event.

Results

HTTP Method	Route	Description	Role Required	Request Body (if any)	Expected Response
GET	`/api/events/{eventId}/results`	Returns the published results for participants in a specific event.	Any (logged in)	None	200 OK - event results. 401 Unauthorized - not authenticated. 404 Not Found - event does not exist.
GET	`/api/results/me`	Returns the authenticated Participant's personal race results across completed events.		None	200 OK - participant's results. 401 Unauthorized - not authenticated
GET	`/api/results/{resultId}`	Returns details of a specific race result	.	None	200 OK - result details. 401 Unauthorized - not authenticated. 403

					Forbidden - user is not authorised to view it. 404 Not Found - result does not exist.
POST	/api/enrolments/{enrollmentId}/result	Captures the finish time, finishing position, and result status for an enrolled Participant after the event	Organiser	{ finishTime, position, resultStatus }	201 Created - result recorded. 400 Bad Request - invalid result data. 401 Unauthorized - not authenticated. 403 Forbidden - organiser does not own the event. 404 Not Found - enrolment does not exist. 409 Conflict - result already exists.
PUT	/api/results/{resultId}	Updates an existing race result, such as	Organiser	{ finishTime, position, resultStatus }	200 OK - result updated. 400 Bad Request - invalid

		correcting the finish time, position, or status.			data. 401 Unauthorized - not authenticated. 403 Forbidden - organiser does not own the event. 404 Not Found - result does not exist.
DELETE	`/api/results/{resultId}`	Removes an incorrect result from the system.	Organiser	None	204 No Content - result deleted. 401 Unauthorized - not authenticated. 403 Forbidden - organiser does not own the event. 404 Not Found - result does not exist.

Enrollments

HTTP Method	Route	Description	Role required	Request Body (if any)	Expected Response

GET	/api/events/{eventId}/enrolments	Returns all participants enrolled in a specific event so the Organiser can view enrolments and statuses.	Organiser	None	200 OK - list of enrolments. 401 Unauthorized - not authenticated. 403 Forbidden - organiser does not own the event. 404 Not Found - event does not exist.
POST	/api/events/{eventId}/enrolments	Enrols the currently authenticated Participant in an event and records the selected category.	Participant	{ categoryId }	201 Created - enrolment created. 400 Bad Request - invalid category/event. 401 Unauthorized - not authenticated. 403 Forbidden - user is not a Participant. 404 Not Found - event or category does not exist. 409 Conflict - participant is already enrolled.

GET	/api/enrolments/me	Returns all events in which the authenticated Participant is enrolled, including enrolment status.	Participant	None	200 OK - participant's enrolments. 401 Unauthorized - not authenticated.
GET	/api/enrolments/{enrollmentId}	Returns details of a specific enrolment for an authorised user.	Any (logged in)	None	200 OK - enrolment details. 401 Unauthorized - not authenticated. 403 Forbidden - user is not authorised to view it. 404 Not Found - enrolment does not exist.
DELETE	/api/enrolments/{enrollmentId}	Withdraws the authenticated Participant from an event enrolment.	Participant	None	204 No Content - enrolment withdrawn. 401 Unauthorized - not authenticated. 403 Forbidden - enrolment does not belong to

					participant . 404 Not Found - enrolment does not exist. 409 Conflict - withdrawal is not allowed.

References

PostMan Team, 2025. What is an API endpoint? [Online], Available at << https://blog.postman.com >> [Accessed 31 August 2026]

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